

Cheetah.

一. 摆腿控制 (swing leg)

$$\tau_i = J_i^T [K_p ({}^B P_{i,ref} - {}^B P_i) + K_d ({}^B V_{i,ref} - {}^B V_i)] +$$

$\tau_{i,fb}$ 反馈控制.

$$J_i^T \Lambda_i ({}^B d_{i,ref} - J_i \dot{q}) + C_i \dot{q}_i + G_i$$

$\tau_{i,ff}$ 前馈项

① 反馈控制

$K_p(\sim) + K_d(\sim) = f_d$ 可看作一种力. $J^T f_d$ 转成力矩.

② 前馈控制.

$$\tau = M \ddot{q} + C \dot{q} + G$$

$$\begin{cases} V = J \dot{q} \Rightarrow \dot{V} = \dot{J} \dot{q} + J \ddot{q} \end{cases}$$

$$\Rightarrow \tau = J^T (J^T)^{-1} M J^{-1} (\dot{V} - \dot{J} \dot{q}) + C \dot{q} + G$$

$$= J^T \Lambda (\dot{V} - \dot{J} \dot{q}) + C \dot{q} + G$$

$\downarrow$
投影 $a_{i,ref}$ 这里没太想明白,

③ 为保证大范围高增益稳定性.

$$k_{p,i} = w_i^2 \wedge i_{ii}$$

二. MPC 接触腿控制 (接触力).

1) 状态变量 $x = [\Theta^T \quad p^T \quad \omega^T \quad \dot{\psi}^T]^T$

其中 Θ 为 ZYX 欧拉角 (先 Z 转 ψ , Y 转 θ , X 转 ϕ)

1) 状态方程.

① 根据欧拉角定义

$${}_1\omega = e_3\dot{\psi} + {}_1e_2'\dot{\theta} + {}_1e_1'\dot{\phi}$$

$$= e_3\dot{\psi} + R_z(\psi)e_2\dot{\theta} + R_z(\psi)R_y(\theta)e_3\dot{\phi}$$

$$= \begin{pmatrix} R_z(\psi)R_y(\theta)e_1 & R_z(\psi)e_1 & e_3 \end{pmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \end{bmatrix}$$

$\dot{\psi}$

$$= \begin{pmatrix} \cos(\psi) \cos\theta & -\sin(\psi) & 0 \\ \sin(\psi) \cos\theta & \cos(\psi) & 0 \\ -\sin\theta & 0 & 1 \end{pmatrix} \quad (1)$$

$$\Rightarrow (4) = \begin{pmatrix} \frac{\cos(\psi)}{\cos\theta} & \frac{\sin(\psi)}{\cos\theta} & 0 \\ -\sin(\psi) & \cos(\psi) & 0 \\ \cos(\psi) \tan\theta & \sin(\psi) \tan\theta & 1 \end{pmatrix} \omega$$

$$\approx I_z \dot{\psi} \omega$$

② 力学方程

$$\sum f_i = m \ddot{p} \Rightarrow \ddot{p} = \frac{1}{m} [I \ I \ \dots \ I] \begin{pmatrix} f_1 \\ i \\ f_n \end{pmatrix}$$

$$\frac{d}{dt} (I\omega) = I\dot{\omega} + \dot{I}\omega \approx I\dot{\omega} = \sum \tau_i f_i$$

$$\Rightarrow \dot{w} = \begin{bmatrix} I^{-1} [r_1] & \dots & I^{-1} [r_n] \end{bmatrix} \begin{bmatrix} f_1 \\ \vdots \\ f_n \end{bmatrix}$$

其中 $L = R_B L R^T$

注：这是质心的力在不同子下的表示，把不同子下的力方程

④ 汇总了得

$$\dot{x} = \begin{bmatrix} 0_3 & 0_3 & R_2(\psi) & 0_3 \\ 0_3 & 0_3 & 0_3 & I_3 \\ 0_3 & 0_3 & 0_3 & 0_3 \\ 0_3 & 0_3 & 0_3 & 0_3 \end{bmatrix} \begin{bmatrix} \textcircled{4} \\ p \\ w \\ p \end{bmatrix}_{12 \times 1}$$

12×12

$$+ \begin{bmatrix} 0_3 & \dots & 0_3 \\ 0_3 & \dots & 0_3 \\ I^{-1} [r_1] & \dots & I^{-1} [r_n] \\ I_3/m & \dots & I_3/m \end{bmatrix} \begin{bmatrix} f_1 \\ \vdots \\ f_n \end{bmatrix}_{3 \times 1} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ g \end{bmatrix}_{12 \times 1}$$

12×31 3×1 12×1

$$\dot{x} = Ax + Bu + y$$

↓ 离散化

$$x_{k+1} = A_k x_k + B_k u_k + y$$

$$\text{其中 } A_k = I + A \Delta T$$

$$B_k = B \Delta T$$

($\Rightarrow$) MPC 问题

$$\min_{x, f} \sum_{k=0}^m \|x(k+1) - x^{\text{ref}}_{k+1}\|_Q + \|f_k\|_R$$

$$\text{s.t. } x_{k+1} = A_k x_k + B_k u_k + y$$

$$|f_x| \leq \mu f_z, \quad |f_y| \leq \mu f_z$$

三. whole body control (WBC)

1. 速度指令计算公式

已知现有 n 个任务, 从优先级由高至低排序为 $T_1(J_1, \dot{x}_1)$, $T_2(J_2, \dot{x}_2)$, ..., $T_n(J_n, \dot{x}_n)$. 求解控制指令 $\dot{q}$, 要求在满足可以完成高优先级任务的前提下再尽量满足低优先级任务。

对于 1 任务 $J_1 \dot{q} = \dot{x}_1 \Rightarrow \tilde{\dot{q}}_1 = J_1^+ \dot{x}_1$

由于 1 的优先级最高, 控制指令应首先完成 $\tilde{\dot{q}}_1$, 则

$\dot{q}_{cmd} = \tilde{\dot{q}}_1$. 对于任务 2, 不干扰任务 1 的完成。则应得。

$\dot{q}_{cmd} = \tilde{\dot{q}}_1 + \tilde{\dot{q}}_2$, 且 $J_1 \tilde{\dot{q}}_2 = 0$, 亦即 $\tilde{\dot{q}}_2$ 位于 J_1 的零空间。

由此可得 ($\dot{q}_{cmd}$ 表示包含所有 n 个任务的控制指令)

$$\tilde{\dot{q}}_2 = \text{Null}(J_1) = \underbrace{(I - J_1^+ J_1)}_{N_1} \dot{z}_2$$

投向了 J_1 零空间的投影解 N_1

也可看作 N_1 的像空间

现在 1 的优先级任务 2 已经完成了, 下来要解决使控制

指定位置满足2.

$$J_2 \dot{q}_{cmd} = J_2 \tilde{q}_1 + J_2 N_1 \dot{z}_2 = \dot{x}_2$$

则可解得 $\dot{z}_2 = (J_2 N_1)^+ (\dot{x}_2 - J_2 \tilde{q}_1)$, 代入 $\dot{q}_{cmd}$

$$\dot{q}_{cmd} = \tilde{q}_1 + N_1 \underbrace{(J_2 N_1)^+ (\dot{x}_2 - J_2 \tilde{q}_1)}_{\text{这里 } -J_2 \tilde{q}_1 \text{ 表示从整个 } \dot{x}_2 \text{ 中把由 } \tilde{q}_1 \text{ 约束着给定成的扣出去}}$$

这里 $-J_2 \tilde{q}_1$ 表示从整个 $\dot{x}_2$ 中把由 $\tilde{q}_1$ 约束着给定成的扣出去

化简, 引理: 若 A 为投影阵, 则 $A(BA)^+ = (BA)^+$

且定义 $J_{21} = J_2 N_1$, 则(1)可化为

$$\dot{q}_{cmd} = \tilde{q}_1 + J_{21}^+ (\dot{x}_2 - J_2 \tilde{q}_1) \quad (2)$$

验证与理解, $J_1 \dot{q}_{cmd} = \underbrace{J_1 J_1^+}_{=I} \dot{x}_1 + \underbrace{J_1 N_1}_{=0} \sim$

这里 $J_1 J_1^+ = I$, 若 J_1 行满秩 (有冗余自由度) 时 J_1^+ 为其右逆

此时 $J_1 J_1^+ = I$, 若 J_1 列满秩或行列都不满秩 (欠驱动)

$J_1 J_1^+ \neq I$, 但是最接近 I 的阵了。这个时候没办法,

因为本来就没有列满秩求一个最小二乘解

$$J_2 \dot{q}_{cmd}^2 = J_1 \tilde{q}_1 + \underbrace{J_2 N_1 (J_1 N_1)^+}_{\text{投影矩阵}} (\dot{x}_2 - J_1 \tilde{q}_1)$$

若 $J_1 N_1$ 的冗余度够用, 则记为 I.

此时 $J_2 \dot{q}_{cmd}^2 = \dot{x}_2$, 冗余度不够用,

那没办法 $J_2 \dot{q}_{cmd}^2 \neq \dot{x}_2$, 误差为左
接近 $\dot{x}_2$

现在在 $\dot{q}_{cmd}^2$ 的基础上, 推回到 $\dot{q}_{cmd}^3$. Let's go!!!

任务三: 可以影响到任务一、二。任务一、二会顺带影响到三。这里有两个难点:

难点1: 设置更空间难点

$$\dot{q}_{cmd}^3 = \dot{q}_{cmd}^2 + \tilde{q}_3 = \dot{q}_{cmd}^2 + N_2 \tilde{q}_3$$

其中 N_2 要满足 $J_1 N_2 = 0$ $J_2 N_2 = 0$ 跟旋子有关

析 $J_1 N_2 = 0$, 可设 $N_2 = N_1 N_x$, 再由 $J_2 N_2 = (J_2 N_1) N_x = 0$

可得 N_x 为 $(J_2 N_1)$ 更空间, 即 $N_2 = N_1 (I - J_{21}^+ J_{21})$

难点2: 通断迭代

$$\text{求 } \dot{\tilde{q}}_1 \text{ 时 } \underline{\dot{\tilde{q}}_1} = \underline{J_1^+ \dot{x}} + \underline{(I - J_1^+ J_1) \dot{z}_1}$$

特解 通解

↓

$$\text{求 } \dot{z}_1 \text{ 时 } \dot{z}_1 = \underline{J_{21}^+ (\dot{x}_2 - J_1 \dot{\tilde{q}}_1)} + \underline{(I - J_{21}^+ J_{21}) \dot{z}_2}$$

特解 通解

↓

回代 $\dot{q}_{cm4}^2$ 与 $\dot{q}_{cm4}^3$

$$\dot{q}_{cm4}^3 = \tilde{\dot{q}}_1 + J_{21}^+ (-) + \underline{N_1 N_{21} \dot{z}_1}$$

与前一一致，记为 N_2

OK, 现在已经有 $\dot{q}_{cm4}^3$ 的表达式, 目标就是搞出 $\dot{z}_3$ 。原理还是和上述类似, $J_3 \dot{q}_{cm4}^3$ 要尽量接近 $\dot{x}_3$

$$J_3 \dot{q}_{cm4}^3 = J_3 \dot{q}_{cm4}^2 + J_3 N_2 \dot{z}_1 = \dot{x}_3$$

$$\Rightarrow \dot{z}_3 = (J_3 N_2)^+ (\dot{x}_3 - J_3 \dot{q}_{cm4}^2)$$

$$\text{因此 } \dot{q}_{cmd}^3 = \dot{q}_{cmd}^2 + N_2 (J_3 N_2)^+ (\dot{x}_3 - J_3 \dot{q}_{cmd}^2)$$

$$\stackrel{4}{(J_3 N_1)^+} = (J_{3|2})^+$$

通式总结

$$\begin{aligned} \dot{q}_{cmd}^1 &= J_1^+ \dot{x}_1 \\ &= \dot{q}_{cmd}^0 + J_{1|0}^+ (\dot{x}_1 - J_1 \dot{q}_{cmd}^0) \\ J_{1|0} &= J_1 N_0 \end{aligned}$$

$$\begin{aligned} \dot{q}_{cmd}^2 &= \dot{q}_{cmd}^1 + J_{2|1}^+ (\dot{x}_2 - J_2 \dot{q}_{cmd}^1) \\ &= \dot{q}_{cmd}^1 + (J_2 N_1)^+ (\dot{x}_2 - J_2 \dot{q}_{cmd}^1) \\ N_1 &= N_0 N_{1|0} \end{aligned}$$

$$\begin{aligned} \dot{q}_{cmd}^3 &= \dot{q}_{cmd}^2 + J_{3|2}^+ (\dot{x}_3 - J_3 \dot{q}_{cmd}^2) \\ &= \dot{q}_{cmd}^2 + (J_3 N_2)^+ (\sim) \\ N_2 &= N_1 N_{2|1} = N_0 N_{1|0} N_{2|1} \end{aligned}$$

这里因为初始 $\dot{q}^1$ 也是全世入的求，要引入一些初始值

$$\dot{q}_i = \dot{q}_{i-1} + J_{i|i-1}^T (x_i - J_i \dot{q}_{i-1})$$

$$\text{其中 } J_{i|i-1} = J_i N_{i-1}$$

$$N_{i-1} = N_0 N_{1|0} \cdots N_{i-1|i-2} = N_{i-2} N_{i-1|i-2}$$

$$N_{i|i-1} = (I - J_{i|i-1}^T J_{i|i-1})$$

$$N_0 = I - J_c^T J_c$$

$$\dot{q}_0 = 0$$

对于 N_0 .

不管什么位置，都要保证接触脚不离地。

$$\text{则有 } 0 = \dot{x} = J_c \dot{q}_0 \Rightarrow \dot{q}_0 \text{ 位于 } \text{null}(J_c)$$

$$\text{因此就得到 } N_0 = I - J_c^T J_c$$

伪代码:

$$q = 0$$

$$N = I - J_c^+ J_c$$

$$J_{tmp} = 0$$

for $i = 1 : n$.

$$N_{tmp} = (I - J_{tmp}^+ J_{tmp})$$

$$N = N N_{tmp}$$

$$J_{tmp} = J_i N$$

$$q = q + J_{tmp}^+ (\dot{x}_i - J_i q)$$

end for.

(二) 位置指令计算.

$$\Delta q_i = \Delta q_{i-1} + J_{i|i-1}^+ \underbrace{(\dot{x}_i^{des} - \dot{x}_i - J_i \Delta q_{i-1})}_{e_i}$$

该定义和速度指令公式定义一样, 因为 Δq 在时间间隔

隔比较短的情况下就近似为 $\dot{q}$ 了, 同理 $e_i \approx \dot{x}_i$ 。这时直

接有因、有果关系

如左图所示 (a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l) (m) (n) (o) (p) (q) (r) (s) (t) (u) (v) (w) (x) (y) (z) (aa) (ab) (ac) (ad) (ae) (af) (ag) (ah) (ai) (aj) (ak) (al) (am) (an) (ao) (ap) (aq) (ar) (as) (at) (au) (av) (aw) (ax) (ay) (az) (ba) (bb) (bc) (bd) (be) (bf) (bg) (bh) (bi) (bj) (bk) (bl) (bm) (bn) (bo) (bp) (bq) (br) (bs) (bt) (bu) (bv) (bw) (bx) (by) (bz) (ca) (cb) (cc) (cd) (ce) (cf) (cg) (ch) (ci) (cj) (ck) (cl) (cm) (cn) (co) (cp) (cq) (cr) (cs) (ct) (cu) (cv) (cw) (cx) (cy) (cz) (da) (db) (dc) (dd) (de) (df) (dg) (dh) (di) (dj) (dk) (dl) (dm) (dn) (do) (dp) (dq) (dr) (ds) (dt) (du) (dv) (dw) (dx) (dy) (dz) (ea) (eb) (ec) (ed) (ee) (ef) (eg) (eh) (ei) (ej) (ek) (el) (em) (en) (eo) (ep) (eq) (er) (es) (et) (eu) (ev) (ew) (ex) (ey) (ez) (fa) (fb) (fc) (fd) (fe) (ff) (fg) (fh) (fi) (fj) (fk) (fl) (fm) (fn) (fo) (fp) (fq) (fr) (fs) (ft) (fu) (fv) (fw) (fx) (fy) (fz) (ga) (gb) (gc) (gd) (ge) (gf) (gg) (gh) (gi) (gj) (gk) (gl) (gm) (gn) (go) (gp) (gq) (gr) (gs) (gt) (gu) (gv) 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$$q = q_i + \Delta q_i$$

(三) 加速度约束计算.

加权约束 / 动态-负载约束: 可看作约束的一种推广, 求 $\min \| \dot{q} \|_M^2 = \min \dot{q}^T M \dot{q}$, 加权约束已最小化 $\| \dot{q} \|_M^2$, M 为一正定矩阵.

$$\min_{\dot{q}} \frac{1}{2} \| \dot{q} \|_M^2 = \min_{\dot{q}} \frac{1}{2} \dot{q}^T M \dot{q}$$

$$\text{s.t. } J\dot{q} = \dot{x}$$

解: 拉格朗日函数 $L = \frac{1}{2} \dot{q}^T M \dot{q} + \lambda^T (\dot{x} - J\dot{q})$

$$\begin{cases} \frac{\partial L}{\partial \dot{q}} = M \dot{q} - J^T \lambda = 0 \\ \frac{\partial L}{\partial \lambda} = \dot{x} - J\dot{q} = 0 \end{cases}$$

解得

$$\begin{cases} \dot{q} = M^{-1} J^T (J M^{-1} J^T)^{-1} \dot{x} \\ \lambda = (J M^{-1} J^T)^{-1} \dot{x} \end{cases}$$

定义约束逆为 $\bar{J}/J^{M+} = M^{-1} J^T (J M^{-1} J^T)^{-1}$

可以证 $J \bar{J} J = J$, $\bar{J}$ 为 J 义逆阵, 但不唯一-彭罗斯逆。由 $J \bar{J} J = J$ 可得 $J(I - \bar{J} J) = 0$, 即 $(I - \bar{J} J)$ 的列依旧在 $M(J)$ 的列空, 但不一定是零矩阵。

由 $\dot{x} = J \dot{q}$ 两侧对时间求导得

$$\dot{\dot{x}}_1 = \dot{J}_1 \dot{q}_1 + J_1 \ddot{q}_1$$

$$\dot{q}_1 = \bar{J}_1 (\ddot{x}_1 - \dot{J}_1 \dot{q}_1) + (I - \bar{J}_1 J_1) \ddot{z}_2$$

|| 右端是已知, 基础上,
|| 左端满足约束

$$J_2 \ddot{q}_{2nd} = J_2 \dot{q}_{2nd} + J_2 M_1 \ddot{z}_2 = \ddot{x}_2 - \dot{J}_2 \dot{q}_2$$

||

$$\ddot{z}_2 = \overline{J_2} N_1 (\ddot{x}_2 - \overline{J_2} \dot{q}_2 - \overline{J_2} \dot{q}_{cmd})$$

$$\text{因此 } \ddot{q}_{cmd}^2 = \dot{q}_{cmd}^1 + N_1 \overline{J_2} N_1 (\ddot{x}_2 - \overline{J_2} \dot{q}_2 - \overline{J_2} \dot{q}_{cmd}^1)$$

$$(N_1 \overline{J_2} N_1 \doteq \overline{J_2} N_1 \text{ 还存在什么抵消})$$

↓ 递推

$$\ddot{q}_{cmd}^i = \ddot{q}_{cmd}^{i-1} + \overline{J_{i|i-1}} (\ddot{x}_i - \overline{J_i} \dot{q}_i - \overline{J_i} \dot{q}_{cmd}^{i-1})$$

$$\text{其中 } \ddot{x}_i = k_p (x_i^{des} - x_i) + k_d (\dot{x}_i^{des} - \dot{x}_i) + \ddot{x}_i^{des}$$

带有反馈。这也是难度程度跟其它指令不同的地方

$$J_{i|pre} = J_i N_{i-1}$$

$$N_i = N_0 N_{1|0} N_{2|1} \dots N_{i|i-1} = N_{i-1} N_{i|i-1}$$

$$N_{i|i-1} = I - \overline{J_{i|i-1}} J_{i|i-1}$$

$$N_0 = I - \overline{J_c} J_c / I - J_c^+ J_c$$

$$\ddot{q}_{cmd} = \tau / (I - J_c^+ J_c)$$

对于闭包有如下解释. 由于无论怎么修, 都无法保持平衡

$$\text{则有 } 0 = J_c \dot{q} \Rightarrow 0 = J_c \ddot{q} + \dot{J}_c \dot{q}$$

$$\text{则 } \ddot{q}_{cmd} = \bar{J}_c (-\dot{J}_c \dot{q})$$

(四) 二次规划松弛

$$\min_{\delta f_r, \delta q_f} \delta f_r^T Q_1 \delta f_r + \delta q_f^T Q_2 \delta q_f$$

$$\text{s.t. } S_f (A \ddot{q} + b + g) = S_f J_c^T f_r \quad \text{浮动总力学}$$

$$\ddot{q} = \ddot{q}_{cmd} + \begin{bmatrix} \delta q_f \\ 0 \end{bmatrix} \quad \text{加速度松弛}$$

$$f_r = f_r^{MPC} + \delta f_r \quad \text{反作用力松弛}$$

$$W f_r \geq 0 \quad \text{接触约束}$$

如果 f_r 与 $\ddot{q}_{cmd}$ 满足浮动总力学, 那么 δf_r 与 $\delta q_f = 0$

但对于某些接触不足, 不完全可控情况, 是解对于加

速度指令与驱动力进行松驰操作。否则这会导致
施加 $\ddot{q}_{my}$ 的优先级。

(2) 驱动力矩控制

$$\begin{bmatrix} \tau_f \\ \tau_j \end{bmatrix} = A\ddot{q} + b + g - J_c^T f_r.$$